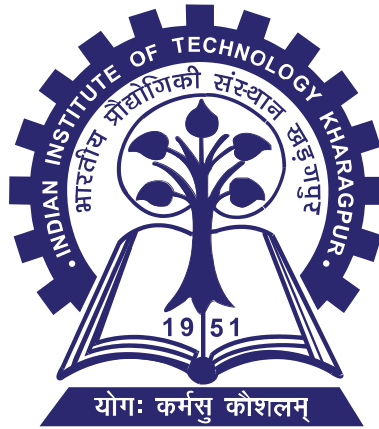




Report On

Mixed-Integer Linear Programming Model for the Optimization of Sawmill Production

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Abstract

The sawmill sector focuses on transforming raw logs into finished lumber through a series of carefully managed steps, ensuring maximum profitability while satisfying market needs. This workflow requires precise coordination of key activities to produce the desired output while making the best use of available resources. By implementing strategic production scheduling that accounts for multiple variables, businesses can enhance efficiency and determine the ideal mix of lumber products. Such optimization helps minimize excess raw material usage, influencing inventory management and supply chain decisions.

In this study, a computational framework was designed to enhance production scheduling, enabling the calculation of the most profitable combination of lumber outputs. The approach leverages mixed-integer linear programming techniques within a profit-maximizing mathematical framework. Key cost components—including production expenses, material expenditures, and procurement outlays—were integral to the model's formulation. Findings indicate that this framework successfully incorporates critical variables, meets market demand, and boosts overall financial performance.

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1 Introduction

The sawmill industry plays a crucial role in processing raw logs into commercially viable sawn timber products. This transformation involves multiple precision operations including initial log breakdown, longitudinal cutting, and final dimensioning processes. The production workflow begins with careful log classification based on physical characteristics, followed by primary sawing operations that yield intermediate products such as cants and flitches. These components undergo further processing to produce finished lumber in various dimensions that meet specific market requirements.

Achieving optimal production efficiency requires careful coordination of several critical factors. The lumber recovery factor, representing the ratio of usable output volume to input log volume, serves as a key performance metric in sawmill operations. Production planning must account for multiple variables including customer-driven product specifications, raw material characteristics, processing methodologies, and equipment capabilities. Effective integration of these elements enables operations to maximize resource utilization while minimizing waste generation.

Recent advancements in operational research have demonstrated the value of mathematical optimization techniques in sawmill production planning. Mixed-integer linear programming approaches have proven particularly effective for addressing complex decision-making scenarios involving discrete production variables and capacity constraints. These methods enable mills to balance competing objectives such as profit maximization, raw material conservation, and demand fulfillment while operating within practical production limitations.

This study presents an optimization framework adapted to real-world sawmill operating conditions, with particular emphasis on incorporating machine capacity constraints into the production planning process. The model development utilizes advanced computational tools specifically designed for solving complex industrial optimization problems. The resulting framework provides practical decision support for production planners seeking to improve operational efficiency and profitability in timber processing operations.

2 Problem Statement

This study investigates production optimization in the Indonesian timber processing industry, particularly focusing on a sawmill operation that utilizes teak logs supplied by KPH Perum Perhutani, a state-run forestry enterprise. The logs are classified by diameter into two types:

All (20–29 cm) and AIII (30 cm and above). Additionally, they are grouped by length into five categories:

- l_1 : 50–90 cm
- l_2 : 100–140 cm
- l_3 : 150–190 cm
- l_4 : 200–240 cm
- l_5 : 250–290 cm

The sawing process consists of three stages: primary cutting using a Band Log Saw, intermediate ripping with a Band Resaw, and final trimming with a Cross-Cut machine. These stages convert raw logs into ten types of sawn timber products. Six of these—Jeblosan Board, Garden Furniture, Flooring, Decking, Longstrip, and Parket Block—are categorized as main products, while the remaining—Parket Stok, Reng, and List—are treated as side products.

Processing is carried out on three machines (PGM 1, PGM 2, and PGM 3), operating 8 hours per day, 6 days a week, over 289 working days annually. The total annual processing capacity across all machines is 10,404 m³. During processing, approximately 80

To improve operational efficiency, a mathematical optimization model is developed by integrating production data, demand requirements, machine constraints, and processing costs into a mixed-integer linear programming (MILP) framework.

3 Notations and Decision Variables

Indices

- d Log diameter (All, AIII)
- l Log length (l1 to l5)
- i Index for main products
- j Index for side products
- p Cutting pattern (CP1, CP2, CP3)

Parameters

- main_sold_i Volume of main product i sold
- side_sold_j Volume of side product j sold
- main_price_i Selling price of main product i
- side_price_j Selling price of side product j
- $\text{log_cost}_{d,l}$ Cost of purchased log
- $\text{stored_log_cost}_{d,l}$ Cost of stored log
- $\text{log_purchased}_{d,l}$ Volume of logs purchased
- $\text{time_per_log}_{d,l,p}$ Time to process logs
- setup_time_p Setup time for pattern p
- $\text{Cap}_{\max}$ Maximum machine capacity
- $T_{\max}$ Total allowable operating time
- F_i, F_j Log proportions for main and side products
- ρ_i, ρ_j Lumber recovery ratios
- BM Big constant for binary constraints

Decision Variables

- $Qa_{d,l}$ Purchased logs used in processing
- $Qt_{d,l}$ Stored logs used in processing
- $Ib_{d,l}$ Stored purchased logs
- $Qp_{d,l,p}$ Total logs processed under cutting pattern p
- $Qpi_{d,l,p}$ Logs processed into main products
- $Qpj_{d,l,p}$ Logs processed into side products

- Pi_i Volume of main product i produced
- Pj_j Volume of side product j produced
- x_p Binary variable for pattern usage

4 Mathematical Formulation

Objective Function

$$\begin{aligned}
 \max Z = & \underbrace{\sum_i \text{main_sold}_i \cdot \text{main_price}_i + \sum_j \text{side_sold}_j \cdot \text{side_price}_j}_{\text{Revenue}} \\
 & - \underbrace{\sum_{d,l} (\log_purchased_{d,l} \cdot \log_cost_{d,l} + Qt_{d,l} \cdot \text{stored_log_cost}_{d,l})}_{\text{Raw Material Cost}} \\
 & - \underbrace{\sum_{d,l,p} Qp_{d,l,p} \cdot Cprd_{d,l,p}}_{\text{Production Cost}}
 \end{aligned}$$

Constraints

Material Balance

$$\begin{aligned}
 \forall d, l : \quad Qa_{d,l} + Ib_{d,l} &= \log_purchased_{d,l} \\
 \forall d, l : \quad Qt_{d,l} &\leq It_{d,l}
 \end{aligned}$$

Product Allocation

$$\begin{aligned}
 \forall d, l, p : \quad Fi \cdot Qp_{d,l,p} &= Qpi_{d,l,p} \\
 \forall d, l, p : \quad Fj \cdot Qp_{d,l,p} &= Qpj_{d,l,p}
 \end{aligned}$$

Production Output

$$\begin{aligned}
 \forall i : \quad \sum_{d,l,p} Qpi_{d,l,p} \cdot \rho_i[d][p][i] &= Pi_i \\
 \forall j : \quad \sum_{d,l,p} Qpj_{d,l,p} \cdot \rho_j[d][p][j] &= Pj_j
 \end{aligned}$$

Log Processing Volume

$$\begin{aligned}\forall d, l : \quad & \sum_p Qp_{d,l,p} = Qt_{d,l} + Qa_{d,l} \\ \forall p : \quad & \sum_{d,l} Qp_{d,l,p} \leq BM \cdot x_p\end{aligned}$$

Machine Operation Constraints

$$\begin{aligned}\sum_{d,l,p} Qp_{d,l,p} \cdot t_{d,l,p} + \sum_p ts_p \cdot x_p & \leq T_{\max} \\ \sum_{d,l,p} Qp_{d,l,p} & \leq \text{Cap}_{\max}\end{aligned}$$

Demand Satisfaction

$$\begin{aligned}\forall i : \quad & Pi_i \geq \text{main_sold}_i = DMi_i \\ \forall j : \quad & Pj_j \geq \text{side_sold}_j = DMj_j\end{aligned}$$

5 Data

The core objective of the production planning process is to optimize both the variety and quantity of sawn timber outputs. These products are categorized into two distinct groups: primary orders, which fulfill specific customer requirements, and secondary orders, comprising non-specified products maintained in inventory for future sales. The inclusion of secondary products serves dual purposes: maximizing raw material utilization and minimizing production waste.

Key operational assumptions governing the production framework include:

- Clear differentiation between customer-specified (primary) and inventory (secondary) product categories
- Standardized product dimensions aligned with both order types
- Sequential processing from raw logs to finished goods without intermediate storage

The manufacturing process utilizes three specialized processing units (PGM 1-3) with a combined annual capacity of 10,404 cubic meters. Operational parameters include:

- Daily operational duration: 8 hours

- Weekly production schedule: 6 working days
- Annual operational period: 289 working days

The production allocation follows a fixed ratio where 80% of processed logs are dedicated to primary orders, with the remaining 20% allocated to secondary products. Supporting operational data, including detailed product specifications and machine parameters, are presented in subsequent tables.

Table 1: The availability of logs (All)

Length	Stored Volume (m ³)	Stored Cost (IDR)	Purchased Volume (m ³)	Purchased Cost (IDR)
I1	6.030	2,328,042	6.030	2,360,388
I2	343.776	2,834,346	343.776	3,291,644
I3	270.993	3,035,619	270.993	3,212,664
I4	143.678	3,044,953	143.678	3,231,399
I5	21.588	3,199,186	21.588	3,453,636

Table 2: The availability of logs (AIII)

Length	Stored Volume (m ³)	Stored Cost (IDR)	Purchased Volume (m ³)	Purchased Cost (IDR)
I1	221.770	4,446,927	421.25	4,456,572
I2	597.530	4,872,151	1253.6	4,884,165
I3	302.880	4,924,974	1279.42	5,057,298
I4	256.970	6,006,590	1580.39	6,261,820
I5	1.740	6,637,329	188.19	6,733,965

Table 3: The demand for sawn timber products

Sawn Timber Products	Volume (m³)	Sawn Timber Products	Volume (m³)
Jeblosan Board	94.653	RST Other	5.248
Garden Furniture	82.729	Parket Block	192.120
Decking	121.711	Parket Stok	145.32
Longstrip	253.759	Reng	176.452
Flooring	1206.444	List	65.69

6 Results and Conclusion

Objective Value (Profit): [1385701535.25862 IDR](#)

Table 4: Purchased Logs Processed (Qa)

Diameter	Length (l)	Volume (m³)
AII	l1	6.03
AII	l2	343.776
AII	l3	270.993
AII	l4	143.678
AII	l5	21.588
AIII	l1	421.25
AIII	l2	1253.6
AIII	l3	1279.42
AIII	l4	1580.39
AIII	l5	188.19

Table 5: Stored Logs Processed (Qt)

Diameter	Length (l)	Volume (m ³)
AII	l1	6.03
AII	l2	309.681923077

Purchased Logs Stored (lb): None listed.

Cutting Patterns Used (x): CP1, CP2

Table 6: Logs Processed by Cutting Pattern (Qp)

Diameter	Length (l)	Pattern	Volume (m ³)
AII	l1	CP1	12.06
AII	l2	CP1	653.457923077
AII	l3	CP1	270.993
AII	l4	CP1	143.678
AII	l5	CP1	21.588
AIII	l1	CP2	421.25
AIII	l2	CP1	1253.6
AIII	l3	CP2	1279.42
AIII	l4	CP1	1322.686923077
AIII	l4	CP2	257.703076923
AIII	l5	CP1	188.19

Table 7: Main Products Production (Pi)

Product	Produced (units)	Demand (units)
Jeblosan_Board	2203.361	94.653
Garden_Furniture	2048.307	82.729
Flooring	1923.779	1206.444
Decking	2141.501	121.711
Skirting	1830.585	5.248
Longstrip	1970.376	253.759
Parket_Block	2063.570	192.12

Table 8: Side Products Production (Pj)

Product	Produced (units)	Demand (units)
Parket_Stok	145.32	145.32
Reng	176.452	176.452
List	122.021	65.69

Table 9: Operation Time Utilization

Metric	Value
Total Time Used (hours)	995.938
Available Time (hours)	2312
Utilization Rate (%)	43.08

Table 10: Raw Material Utilization

Metric	Value
Total Logs Processed (m ³)	5824.627
Total Capacity (m ³)	10404
Utilization Rate (%)	55.98

The solution achieves a high profit of IDR 1.39 billion, efficiently meeting all product demands while utilizing 43.08% of operation time and 55.98% of raw material capacity. The production plan is cost-effective and scalable, with room for increased utilization.

7 Cplex Code

7.1 .mod file

```

1  /*****
2   * Mixed-Integer Linear Programming Model
3   * for Sawn Timber Production Planning (.mod)
4   *****/
5
6  // Sets
7  {string} Diameter = ...; // Diameters of logs
8  {string} Length = ...; // Length of logs
9  {string} MainProducts = ...; // Main products
10 {string} SideProducts = ...; // Side products
11 {string} CuttingPattern = ...; // Cutting patterns
12
13 // Parameters
14 // Sawn timber products sold
15 float VFi[MainProducts] = ...;
16 float VFj[SideProducts] = ...;
17
18 // Selling price per product

```

```

19 float PVi[MainProducts] = ...;
20 float PVj[SideProducts] = ...;
21
22 // Stored logs availability and cost
23 float It[Diameter][Length] = ...;
24 float Crmp[Diameter][Length] = ...;
25
26 // Purchased logs cost and availability
27 float Crm[Diameter][Length] = ...;
28 float Qb[Diameter][Length] = ...;
29
30 // Production cost for each cutting pattern
31 float Cprd[Diameter][Length][CuttingPattern] = ...;
32
33 // Proportion of logs processing
34 float Fi = ...; // Main product proportion
35 float Fj = ...; // Side product proportion
36
37 // Conversion factor (lumber recovery ratio)
38 float rho_i[Diameter][CuttingPattern][MainProducts] = ...;
39 float rho_j[Diameter][CuttingPattern][SideProducts] = ...;
40
41 // Demand for products
42 float DMi[MainProducts] = ...;
43 float DMj[SideProducts] = ...;
44
45 // Operating time for processing logs
46 float t[Diameter][Length][CuttingPattern] = ...;
47
48 // Setup time for cutting pattern
49 float ts[CuttingPattern] = ...;
50
51 // Maximum operation time (8 hours per day * 289 days)
52 float Tmax = ...;
53

```

```

54 // Maximum production capacity
55 float Capmax = ...;
56
57 // Big constant for the model
58 float BM = ...;
59
60 // Decision Variables
61 dvar float+ Qa[Diameter][Length];      // Purchased logs
        processed
62 dvar float+ Qt[Diameter][Length];      // Stored logs processed
63 dvar float+ Ib[Diameter][Length];      // Purchased logs stored
64 dvar float+ Qp[Diameter][Length][CuttingPattern];    // Total
        logs processed
65 dvar float+ Qpi[Diameter][Length][CuttingPattern];    // Logs
        for main products
66 dvar float+ Qpj[Diameter][Length][CuttingPattern];    // Logs
        for side products
67 dvar float+ Pi[MainProducts];          // Production of main
        products
68 dvar float+ Pj[SideProducts];          // Production of side
        products
69 dvar boolean x[CuttingPattern];        // Cutting pattern binary
        variable
70
71 // Objective Function: Maximize profit
72 maximize
73     // Sales revenue
74     (sum(i in MainProducts) VFi[i] * PVi[i] + sum(j in
        SideProducts) VFj[j] * PVj[j]) -
75     // Raw material cost
76     (sum(d in Diameter, l in Length) (Qb[d][l] * Crm[d][l] + Qt[d
        ][l] * Crmp[d][l])) -
77     // Production cost
78     (sum(d in Diameter, l in Length, p in CuttingPattern) Qp[d][l
        ][p] * Cprd[d][l][p]);

```

```

79
80 // Constraints
81 subject to {
82     // Raw material availability constraints
83     forall(d in Diameter, l in Length)
84         Qb[d][l] == Qa[d][l] + Ib[d][l];
85
86     forall(d in Diameter, l in Length)
87         It[d][l] >= Qt[d][l];
88
89     // Log processing constraints
90     forall(d in Diameter, l in Length, p in CuttingPattern)
91         Fi * Qp[d][l][p] == Qpi[d][l][p];
92
93     forall(d in Diameter, l in Length, p in CuttingPattern)
94         Fj * Qp[d][l][p] == Qpj[d][l][p];
95
96     // Production of main products
97     forall(i in MainProducts)
98         sum(d in Diameter, l in Length, p in CuttingPattern) Qpi[d][
99             l][p] * rho_i[d][p][i] == Pi[i];
100
101     // Production of side products
102     forall(j in SideProducts)
103         sum(d in Diameter, l in Length, p in CuttingPattern) Qpj[d][
104             l][p] * rho_j[d][p][j] == Pj[j];
105
106     // Total logs processed
107     forall(d in Diameter, l in Length)
108         sum(p in CuttingPattern) Qp[d][l][p] == Qt[d][l] + Qa[d][l];
109
110     // Cutting pattern usage
111     forall(p in CuttingPattern)
112         sum(d in Diameter, l in Length) Qp[d][l][p] <= BM * x[p];

```

```

112 // Operating time constraint
113 sum(d in Diameter, l in Length, p in CuttingPattern) Qp[d][l][
    p] * t[d][l][p] + sum(p in CuttingPattern) ts[p] * x[p] <=
    Tmax;
114
115 // Production capacity constraint
116 sum(d in Diameter, l in Length, p in CuttingPattern) Qp[d][l][
    p] <= Capmax;
117
118 // Production must satisfy demand
119 forall(i in MainProducts)
120     Pi[i] >= VFi[i];
121
122 forall(j in SideProducts)
123     Pj[j] >= VFj[j];
124
125 forall(i in MainProducts)
126     VFi[i] == Dmi[i];
127
128 forall(j in SideProducts)
129     VFj[j] == DMj[j];
130 }

```

7.2 .dat file

```

1
2 /*****
3  * Mixed-Integer Linear Programming Model
4  * for Sawn Timber Production Planning (.dat)
5  *****/
6 // Sets
7 Diameter = {"AII", "AIII"};
8 Length = {"11", "12", "13", "14", "15"};
9 MainProducts = {"Jeblosan_Board", "Garden_Furniture", "Flooring"
    , "Decking", "Skirting", "Longstrip", "Parquet_Block"};

```

```

10 SideProducts = {"Parket_Stok", "Reng", "List"};
11 CuttingPattern = {"CP1", "CP2", "CP3"};
12
13 // Parameters
14 // Sawn timber products sold (demand)
15 VFi = [94.653, 82.729, 1206.444, 121.711, 5.248, 253.759,
16        192.120];
17
18 VFj = [145.32, 176.452, 65.69];
19
20 // Selling price per product (IDR/m^3)
21 PVi = [12500000, 14000000, 13500000, 12000000, 11500000,
22        13000000, 12750000];
23 PVj = [15000000, 10000000, 90000000];
24
25 // Stored logs availability and cost
26 It = [
27     [6.030, 343.776, 270.993, 143.678, 21.588], // AII
28     [221.770, 597.530, 302.880, 256.970, 1.740] // AIII
29 ];
30 Crmp = [
31     [2328042, 2834346, 3035619, 3044953, 3199186], // AII
32     [4446927, 4872151, 4924974, 6006590, 6637329] // AIII
33 ];
34
35 // Purchased logs cost and availability
36 Crm = [
37     [2360388, 3291644, 3212664, 3231399, 3453636], // AII
38     [4456572, 4884165, 5057298, 6261820, 6733965] // AIII
39 ];
40 Qb = [
41     [6.030, 343.776, 270.993, 143.678, 21.588], // AII
42     [421.25, 1253.6, 1279.42, 1580.39, 188.19] // AIII
43 ];
44
45 // Production cost for each cutting pattern

```

```

43 Cprd = [
44     // AII
45     [
46         [800000, 850000, 820000], // 11
47         [830000, 880000, 850000], // 12
48         [850000, 900000, 870000], // 13
49         [880000, 930000, 900000], // 14
50         [900000, 950000, 920000] // 15
51     ],
52     // AIII
53     [
54         [850000, 900000, 870000], // 11
55         [880000, 930000, 900000], // 12
56         [900000, 950000, 920000], // 13
57         [930000, 980000, 950000], // 14
58         [950000, 1000000, 970000] // 15
59     ]
60 ];
61
62 // Proportion of logs processing
63 Fi = 0.8; // Main product proportion
64 Fj = 0.2; // Side product proportion
65
66 // Conversion factor (lumber recovery ratio)
67 rho_i = [
68     // AII
69     [
70         [0.46, 0.42, 0.40, 0.44, 0.38, 0.41, 0.43], // CP1
71         [0.45, 0.43, 0.39, 0.45, 0.37, 0.40, 0.42], // CP2
72         [0.44, 0.41, 0.38, 0.43, 0.36, 0.39, 0.41] // CP3
73     ],
74     // AIII
75     [
76         [0.48, 0.44, 0.42, 0.46, 0.40, 0.43, 0.45], // CP1
77         [0.47, 0.45, 0.41, 0.47, 0.39, 0.42, 0.44], // CP2

```

```

78     [0.46, 0.43, 0.40, 0.45, 0.38, 0.41, 0.43]    // CP3
79 ]
80 ];
81 rho_j = [
82     // AII
83     [
84         [0.12, 0.14, 0.10],    // CP1
85         [0.11, 0.15, 0.09],    // CP2
86         [0.10, 0.13, 0.08]    // CP3
87     ],
88     // AIII
89     [
90         [0.13, 0.15, 0.11],    // CP1
91         [0.12, 0.16, 0.10],    // CP2
92         [0.11, 0.14, 0.09]    // CP3
93     ]
94 ];
95
96 // Demand for products
97 DMi = [94.653, 82.729, 1206.444, 121.711, 5.248, 253.759,
98       192.120];
99
100 // Operating time for processing logs
101 t = [
102     // AII
103     [
104         [0.12, 0.14, 0.13],    // 11
105         [0.13, 0.15, 0.14],    // 12
106         [0.15, 0.17, 0.16],    // 13
107         [0.17, 0.19, 0.18],    // 14
108         [0.19, 0.21, 0.20]    // 15
109     ],
110     // AIII
111     [

```

```
112     [0.13, 0.15, 0.14], // 11
113     [0.15, 0.17, 0.16], // 12
114     [0.17, 0.19, 0.18], // 13
115     [0.19, 0.21, 0.20], // 14
116     [0.21, 0.23, 0.22] // 15
117 ]
118 ];
119
120 // Setup time for cutting pattern
121 ts = [0.5, 0.6, 0.55];
122
123 // Maximum operation time (8 hours per day * 289 days)
124 Tmax = 2312;
125
126 // Maximum production capacity
127 Capmax = 10404;
128
129 // Big constant for the model
130 BM = 1000000000000000;
```

References

- [1] Octavia Riskadayanti, Muhammad Hisjam, Y Yuniaristanto. *Mixed-Integer Linear Programming Model for Production Planning*
- [2] SergioMaturana, EnzoPizani. *Schedulingandproductionplanningfora-sawmill:acomparisonofa mathematicalmodelversusaheuristic*
- [3] Rahman, Asif & Yella, Siril & Dougherty, Mark. (2014). *Simulation and Optimization Techniques for Sawmill Yard Operations—A Literature Review. Journal of Intelligent Learning Systems and Applications.*